Ship Shape: Practice Test 1

Time: 35 minutes (the part of the 50-minute event after the 15-minute build). Work with your partner and write one set of answers. Bring Category C indirect-vent goggles, scissors, a ruler, and pencils; a basic calculator, protractor, and tape measure are optional. Keep working on the test while you wait, and stop only when you are called to test your barge. There are 40 questions worth half a point each, so the test is out of 20 points, like the real one.

Partner names: _____ and _____ Score: _____

Part 1: Buoyancy (the upward push)

1. Which way does buoyant force push on an object in water?

◯ A Down ◯ B Up ◯ C Sideways only ◯ D It does not push at all

2. A toy boat floats still on a pond. Which two forces are acting on it?

◯ A Gravity pulling down and buoyant force pushing up ◯ B Only gravity ◯ C Only buoyant force
◯ D Magnetism and friction

3. A toy boat has a mass of 40 g and floats still on the water. How many grams does the water's upward push hold up?

_____

4. You hold a beach ball under water and then let go. What happens and why?

◯ A It stays where it is ◯ B It shoots up, because the buoyant force is bigger than its weight
◯ C It sinks slowly ◯ D It sinks fast

5. When you push an empty capped bottle deeper into water, it pushes back harder. Why?

_____

_____

6. A rock sinks to the bottom of a bucket. Is the water still pushing up on it?

◯ A Yes, but the push is smaller than the rock's weight ◯ B No, water only pushes on things that float
◯ C Yes, and the push is bigger than the rock's weight

Part 2: Displacement

7. What does displacement mean?

◯ A Water drying up in the sun ◯ B Water freezing into ice
◯ C Water being pushed aside by an object, so the water level rises ◯ D Water leaking into a boat

8. A measuring cup has 200 mL of water. You drop in a toy dinosaur and it sinks all the way. The water rises to 245 mL. What is the volume of the dinosaur?

_____

9. A marble has a volume of 6 cubic cm. You drop 5 marbles into a cup filled to 100 mL. What will the new water level be?

_____

10. Where should your eyes be when you read the water level in a measuring cup?

_____

11. A wooden block with a mass of 30 g floats in a cup. How much water does it push aside?

_____

12. A steel ball and a glass ball are exactly the same size. Both sink all the way under. Which one pushes aside more water?

◯ A The steel ball, because it is heavier ◯ B The glass ball, because it is lighter
◯ C They push aside the same amount, because they are the same size

Part 3: Archimedes' principle

13. Archimedes was a scientist and inventor who lived long ago. Where was he from?

◯ A Ancient Egypt ◯ B Ancient Greece (the Greek city of Syracuse) ◯ C Ancient China
◯ D Ancient Rome

14. Finish Archimedes' principle: The upward push on an object equals the weight of the _____ it pushes aside.

---

15. A rock with a volume of 25 cubic cm is all the way under water. How much water does it push aside, and how big is the upward push (in grams)?

---

16. That same rock has a mass of 60 g. If you hang it from a scale while it is under water, what will the scale read? Show your math.

---

17. A barge floating with its cargo pushes aside 300 mL of water. What is the mass of the barge plus its cargo?

---

18. As cargo is added, a barge sits lower in the water. Which statement is true?

◯ A It pushes aside more water, so the upward push gets bigger
◯ B It pushes aside less water, so the upward push gets smaller ◯ C The upward push stays the same
◯ D Water stops pushing on it

Part 4: Density and flotation

19. What is density?

20. The density of water is _____ gram(s) per cubic cm.

21. A block of wood has a mass of 50 g and a volume of 100 cubic cm. What is its density? Will it float or sink in water?

22. A metal cube is 2 cm x 2 cm x 2 cm and has a mass of 64 g. What is its volume? What is its density? Will it float or sink?

23. A plastic toy has a mass of 45 g and a volume of 50 cubic cm. Will it float or sink? Explain.

24. Why do ice cubes float in a glass of water?

◯ A Ice is heavier than water ◯ B Ice is colder than water ◯ C Ice is less dense than liquid water
◯ D Ice has air bubbles only on top

25. An egg sinks in a glass of tap water. After you stir in a lot of salt, the egg floats. Why?

◯ A The salt makes the egg lighter ◯ B Salt water is denser than tap water, so it pushes up harder
◯ C The salt melts the egg shell ◯ D Salt water is less dense than tap water

26. Put these in order from least dense to most dense: steel, water, cork.

27. Maya's 20 g clay ball sinks. She presses it into a bowl shape and it floats. What stayed the same, and what changed?

28. A submarine lets water into its tanks to go down, and pushes the water out with air to come up. Letting water in makes the submarine:

◯ A Denser than the water around it, so it sinks ◯ B Less dense than the water around it, so it sinks
◯ C Smaller, so it floats

Part 5: Barge math and units

Use: volume = length x width x height, and 1 cubic cm of water = 1 mL = 1 g.

29. A barge's cargo area is 14 cm long, 12 cm wide, and has walls 2 cm tall. What is its volume? How many grams of water can it push aside?

30. That barge has a mass of 11 g. What is the most cargo it can hold (in a perfect world)?

31. Which barge can push aside more water: Barge A is 15 cm x 10 cm x 2.5 cm, and Barge B is 12 cm x 12 cm x 2.5 cm? Show your math.

32. 10 hex nuts have a mass of 31.40 g. What is the mass of 1 nut?

33. A barge held 72 nuts. Each nut has a mass of 3.14 g. What mass of cargo did it hold?

34. Your math says your barge can hold 300 g. At practice, your barges held only about 3 out of every 4 grams the math said (75 percent). What is a good estimate?

35. Which estimate is written the right way for the score sheet?

◯ A about 250 g ◯ B 250 g ◯ C 250.00 g ◯ D 25 nuts

Part 6: Rules and scoring

36. Which eye protection must you wear while testing your barge?

◯ A Regular eyeglasses ◯ B Category C indirect-vent goggles ◯ C Standard safety glasses
◯ D Sunglasses

37. A pair's barge has a flat lid of plastic taped over the cargo area with a small hole in it. Which rule does it break?

38. During testing, some water leaks into the barge, but the top edge of the cargo area is still above the water. Can the judge keep adding nuts? Can you scoop the water out?

39. What is the most cargo the judge will load onto any barge?

40. A pair scores 16 test points. Its barge holds 204.50 g, and its estimate was exactly 204.50 g. What is its final score?

31	A: 15 x 10 x 2.5 = 375 cubic cm. B: 12 x 12 x 2.5 = 360 cubic cm. Barge A pushes aside more (375 g).
32	31.40 / 10 = 3.14 g.
33	72 x 3.14 = 226.08 g.
34	300 x 0.75 = 225 g (then round to a whole number of nuts and write it to 0.01 g).
35	C 250.00 g
Part 6: Rules and scoring	
36	B Category C indirect-vent goggles
37	The cargo area must be open at the top for loading. It is Tier 2.
38	Yes, loading keeps going. No, you may not remove any water.
39	500.00 g
40	16 + 204.50 + 5 bonus = 225.50.